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Original Research

The impossibility of mortality evaluation of skin cancer screening in Germany based on health insurance data: a case–control study

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Abstract ***Background:*** The aim of this paper was to perform a mortality evaluation of skin cancer screening (SCS) in Germany using General Local Health Insurance Fund (AOK) data.
Methods: AOK-insured men and women aged 35–99 years who died of cutaneous malignant melanoma in 2015–2016 were identified. Controls were AOK-insured people who survived to the end of each case's year of death. For each case, 10 controls were matched. The SCS history of each individual was reconstructed using the billing codes 01745 and 01746.
Results: In total, 1037 melanoma deaths and 10,370 controls were included. Cumulative SCS prevalence increased among controls over calendar years, as expected (males and females, 2009: 13.5% and 12.5%; 2015: 52.1% and 55.1%). In contrast, among cases, cumulative SCS prevalence was already high in 2009 and did not show a monotonic increase over the years of diagnosis. Of the 1037 melanoma deaths, 224 (21.6%) had at least one SCS settled in the 12 months after diagnosis.

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Discussion: A mortality evaluation with health insurance data alone is not possible because SCS billing codes are not only used for real SCS but also for occasion-related diagnostic work-up of abnormal skin findings. A mortality evaluation with health insurance data requires an individual linking with data of the screening physician and the cancer registries.
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1. Introduction

In August 2008, the biennial skin cancer screening (SCS) for persons aged 35 years and older, financed by the statutory health insurances, was started. It includes targeted medical history and whole body inspection of the entire skin including the hairy head and all intertrigines. Since April 2020, dermoscopy has also been integrated. The SCS can only be performed by contracted physicians of certain specialities who have an authorisation from the responsible Association of Statutory Health Insurance Physicians. It is approved for specialists in general medicine, general internists, general practitioners without a specialist designation, and dermatologists. Before admission to the SCS, physicians must demonstrate successful participation in an 8-h continuing education program certified by the Association of Statutory Health Insurance Physicians [1]. Among the 24.5 million people aged 37 years or more who were insured by the General Local Health Insurance Fund (AOK), 39% had an SCS billing code documented within 2016–2018 [2].

The decisive reason for the introduction of the SCS was the potential benefit by bringing forward the time of diagnosis (lead time) and thus, as expected, a diagnosis in the lower stage. Clear evidence for a reduction in skin cancer mortality or morbidity was lacking at the time of approval of the SCS [3]. Results on the reduction of skin cancer mortality in the SCREEN model project in Schleswig–Holstein, Germany, were published in 2012 [4]. However, various papers questioned the validity of the observed skin cancer mortality reduction of 50% within 2 years [4–8]. A recent study that compared survival between health-insured skin melanoma patients either detected within the screening program or outside the screening program found that it was unable to separate the screening effect from other effects as, e.g., lead time bias [9].

In absolute numbers, cutaneous malignant melanomas contribute most to skin cancer mortality in Germany, which is why the effect of the SCS on the reduction of melanoma mortality in particular is of interest. The evaluation of SCS by means of a randomised trial is ethically not justifiable due to the already existing approval of SCS in Germany. Furthermore, because of the statistically low melanoma mortality of approximately 3.6 per 100,000 person-years (calendar year 2019, nationwide, men and women combined, www.gbe-bund.de, 9/22/2021), a randomised trial would need to enrol an extremely large number of subjects to statistically ensure mortality reduction. For this reason, we sought to conduct a program evaluation of SCS using health insurance data. The aim of this work was to evaluate whether participation in the SCS is associated with a reduction in melanoma mortality using a case–control study design.

2. Methods

The case–control study evaluating melanoma mortality is one of the three projects conducted in a collaborative called EvaSCa "Evaluation of Skin Cancer" funded by the Innovation Committee of the Federal Joint Committee in Germany. Results of the other two projects that deal with the treatment intensity of the following 12 months depending on the detection of a skin cancer (within or outside the screening) and with the health economic consequences (direct costs) are presented elsewhere.

2.1. Case definition

The source population of the case–control study was AOK-insured persons in Germany (24.5 million people in 2015). A random sample of 450,000 AOK-insured individuals was used to identify individuals who received a diagnosis of cutaneous malignant melanoma (ICD-10 [10]: C43) in 2014 or 2015. For 96.8% of cases, there was continuous insurance since 2008, for 0.9% since 2009, for 1.9% since 2010, and for 0.4% since 2011. For all patients, we examined when the diagnosis of C43 first occurred between 2008 and 2015. To avoid SCS history bias, we excluded melanoma patients who had a prior diagnosis of other malignant neoplasm of the skin (ICD-10: C44) before the skin melanoma diagnosis.

The first-ever skin melanoma diagnosis was counted as incident melanoma only if surgical or medical treatment for the melanoma was also coded in the temporal context (12 months before to three months after the diagnosis date). Only cases between the ages of 35 and 99 years were included. We used three criteria that all had to apply to define a skin melanoma death. First, an incident melanoma patient who died in 2015–2016 counted as a case if the time interval between the diagnosis date and the date of death was at least 1 year. We assumed that a time interval between initial diagnosis of

melanoma and death of less than 1 year makes melanoma unrealistic as a cause of death. Second, the time interval between the date of the last melanoma diagnosis before death and the date of death was 1 year or less. Here, we assumed that if a patient died of melanoma, a diagnosis of melanoma (ICD-10C43) would appear in the insured data at least once in the year preceding the death. Third, patients were not allowed to have competing causes of a death in 3 months before death. Application of these three inclusion criteria was intended to reduce the potential misclassification of cases. Increasing the specificity at the expense of sensitivity for outcome assessment, here melanoma death, comes at the expense of the number of cases in the case–control study but results in a lower misclassification bias of the effect estimate compared with increasing sensitivity at the expense of specificity [11]. Following Gagnon *et al.* [12], competing causes of death included the following ICD-10 diagnoses: C15–C26, C34, C45, C46, C50, C53–C54, C61, C64, C67, C80, C92, I50.14, I50.05 N18.5, R57.2, and K72.

2.2. Control definition

Controls were AOK insureds who survived to the end of the year in which the corresponding case died. Sampling of controls was independent of whether or not a control had already suffered melanoma. Ten controls each was matched to cases. Matching criteria were age (5-year groups) at the time of the case's date of death, sex (male, female), and state of residence (16 states). The diagnosis date of the incident melanoma of each case was used to assign a pseudodiagnosis date for the corresponding controls. Further on, the diagnosis date of the cases and the pseudodiagnosis date of the controls are called reference dates.

2.3. HKS history

Insured persons for whom, in the 24 or 3 months preceding the reference date, the billing codes 01745 (screening examination for skin cancer in accordance with Section D. II. of the Cancer Screening Guideline) or 01746 (supplement to fee schedule number 01732 for screening examination for skin cancer in accordance with Section D. II. of the Cancer Screening Guideline) were assigned were considered screened individuals (exposed individuals). In the absence of these billing codes the periods, insured persons were classified as non-screened persons (unexposed persons).

The comorbidity of the insured was derived using the Elixhauser comorbidity index [13] based on ICD-10-coded diagnoses up to 1 year before the reference date. According to Section 303e of the Fifth Book of the German Social Code (SGB), universities, among others, have the right to use health insurance data for research purposes. The data of insured persons were completely anonymised and stored on a server with restricted access and password protection. Data owners had no input into the research project. The project followed the guidelines and recommendations of the German "Good Practice of Secondary Data Analysis" [14]. The Ethics Committee of Essen University Hospital approved the study protocol without objection on November 26, 2019 (reference 19-8981-BO).

3. Statistical methods

The sample size calculation yielded 1024 cases and 4095 controls to have a statistical power 1-β of 80% for a two-sided test with $\alpha = 0.05$ and a minimum odds ratio of 0.80. To evaluate the validity of the billing codes indicating an SCS, the prevalence of an SCS that had ever occurred since 2008 (start of the SCS program), i.e., between the first insured year and the reference date, was determined as a function of the year of the reference date. It was expected that as the reference date increased, this cumulative prevalence would increase because the period of prevalence increases as the reference date increases. Furthermore, we determined the SCS prevalence in the last 3 and 24 months before the reference date.

For controls, we compared the age- and sex-specific SCS prevalences for the 3-year periods 2009–2011 and 2012–2014 with the published prevalence of the total AOK-insured population, which served as the source population [2]. We determined the expected number of melanoma deaths in 2014–2015 within the random sample of AOK insureds using the sex- and age-specific nationwide melanoma mortality rates and the corresponding numbers of AOK-insured years. This extrapolation assumes that melanoma mortality rates as observed nationwide would be so observed among AOK insureds. One study showed that employed persons among AOK-insured persons are on average lower skilled and that the proportion of employed women is lower than nationwide [15]. We then compared the expected numbers of melanoma deaths with the numbers included in the study.

4. Results

A total of 1037 melanoma death cases and 10 times that number of controls (10,370) were included in the study. Because of matching for age, sex, and place of residence (state), these distributions were the same for cases and controls. The median Elixhauser comorbidity index did not differ between cases and controls (Table 1).

Comparison of cumulative SCS prevalence between cases and controls showed that for controls, cumulative HKS prevalence increased with increasing year of reference date. For example, the cumulative SCS prevalence in male and female controls in 2009 was 13.5%

Table 1
Characteristics of 1037 skin melanoma deaths and 10,370 matching controls sampled from the General Local Health Insurance Fund ("Allgemeine Ortskrankenkasse, AOK") in Germany, 2014–2015.

Characteristic	Cases		Controls	
	N	%	N	%
Sex				
Men	541	52.2	5410	52.2
Women	496	47.8	4960	47.8
Age (years) at diagnosis*				
35–49	71	6.8	680	6.6
50–59	97	9.4	940	9.1
60–69	140	13.5	1420	13.7
70–79	355	34.2	3580	34.5
80–89	316	30.5	3310	31.9
90–99	60	5.8	440	4.2
Place of residence (State)				
Schleswig–Holstein	27	2.6	270	2.6
Hamburg	9	0.9	90	0.9
Bremen	4	0.4	40	0.4
Niedersachsen	93	9.0	930	9.0
Nordrhein-Westfalen	159	15.3	1590	15.3
Rheinland-Pfalz	48	4.6	480	4.6
Hessen	73	7.0	730	7.0
Saarland	5	0.5	50	0.5
Baden-Württemberg	119	11.5	1190	11.5
Bayern	202	19.5	2020	19.5
Berlin	30	2.9	300	2.9
Brandenburg	44	4.2	440	4.2
Sachsen	101	9.7	1010	9.7
Sachsen-Anhalt	42	4.1	420	4.1
Mecklenburg-Vorpommern	30	2.9	300	2.9
Thüringen	50	4.8	500	4.8
Elixhauser Comorbity Index (Mean, Median)	1.92	0	0.84	0

All percentages are column percentages; *for controls, the pseudo-diagnosis date were used.

and 12.5%, respectively, and in 2015 was 52.1% and 55.1%, respectively. In contrast, among male and female cases, the cumulative SCS prevalence was already high in 2009 and did not show a monotonic increase over the years of diagnosis. The prevalence of SCS in the last 3 months before pseudodiagnosis date (controls) was low in all years, with values ranging from 3 to 6%. Among cases, this prevalence was 32–46% (men) and 20–34% (women). The SCS prevalence among melanoma deaths that received the first melanoma diagnosis already in 2009, i.e., at most about 1.5 years after the introduction of SCS in Germany, was already 57.4% and 38.0% in men and women, respectively (Table 2).

Comparison of SCS prevalence three and 24 months prior to reference date between cases and controls showed that SCS prevalence was markedly higher in cases than in controls at each age and sex stratum. Overall, 36.4% (men) and 26.8% (women) were billed for SCS in the 3 months before reference date at initial melanoma diagnosis. Even in the 80–89 age group, 34.6% of male cases and 24.4% of female cases, respectively, were billed for an SCS at initial melanoma diagnosis (Table 3). A total of 224 of the 1037 melanoma deaths (21.6%) had at least one SCS billed in the 12 months after diagnosis, predominantly by dermatologists (171 of 224, 76.3%).

Comparison of age- and sex-specific SCS prevalence for the 3-year period 2009–2011 and 2012–2014 between the sampled controls and all AOK-insured persons revealed that the SCS prevalence of the controls relatively closely mirrored the SCS prevalence of the source population, so that the bias of the SCS prevalence comes from the cases (Supplementary Fig. 1). Of the expected 1687 melanoma deaths within the sample, we identified and included 1037 melanoma deaths (61.5%) in the case–control study. The non-inclusion of 650 potential melanoma deaths was due to the application of the three inclusion criteria, which were more frequently not met with increasing age (Supplementary Table 1).

5. Discussion

Our results show that a mortality evaluation of SCS based on insured data alone is not possible in Germany because a valid estimate of screening prevalence in melanoma deaths based on SCS billing codes is not possible. This conclusion must be reached because, as expected, by extending the time interval of a cumulative prevalence (SCS ever since 2008), the cumulative prevalence of SCS must increase. While this expectation was met for controls, which are a reflection of the source population within the matching strata, cases did not show an upward trend in cumulative prevalence over the years of diagnosis but instead showed a very high SCS prevalence already in 2009. Even for a mortality evaluation, where one would rather use the prevalence of an SCS in, e.g., the last 3 or 24 months before reference date, there is a corresponding bias towards a massive overestimation of the screening prevalence. In 21.6% of melanoma deaths, even at least one SCS was billed in the first 12 months after (sic) diagnosis, although "a renewed screening examination for skin cancer … is only possible after the end of the calendar year following the previous examination" [1].

Our study revealed that the sole routine data of health insurances in Germany are not suitable for the mortality evaluation of SCS because the billing codes for SCS are not exclusively used for an occasion-free screening but are obviously also used in the context of occasion-related diagnostic work-up of patients with abnormalities of the skin. An occasion such as the discovery of a conspicuous nevus may have arisen with the patient himself, his partner or also with physicians and other service providers (e.g., physiotherapists) who do not themselves perform the screening. The diagnostic work-up of such abnormal findings does not fall under the concept of screening, in which persons without symptoms are submitted to the screening test, but under the concept of early diagnosis, in which persons with

Table 2
Skin cancer screening prevalence in relation to the year of diagnosis among 1037 skin melanoma deaths and in relation to the year of the pseudodiagnosis among 10,370 controls among General Local Health Insurance Fund ("Allgemeine Ortskrankenkasse, AOK") in Germany, 2014–2015.

Year of (Pseudo) Diagnosis	Men						Women					
	Cases			Controls			Cases			Controls		
	N	n	%	N	n	%	N	n	%	N	n	%
Ever skin cancer screening (*back to 2008*)												
2009	68	39	57.4	680	92	13.5	71	27	38.0	710	89	12.5
2010	72	34	47.2	720	199	27.6	67	38	56.7	670	165	24.6
2011	56	33	58.9	560	192	34.3	63	28	44.4	630	197	31.3
2012	86	45	52.3	860	356	41.4	83	38	45.8	830	305	36.7
2013	104	62	59.6	1040	449	43.2	95	41	43.2	950	414	43.6
2014	116	66	56.9	1160	588	50.7	82	45	54.9	820	382	46.6
2015	39	21	53.8	390	203	52.1	35	20	57.1	350	193	55.1
Skin cancer screening within the recent 24 months *before date of diagnosis**												
2009	68	39	57.4	680	92	13.5	71	27	38.0	710	89	12.5
2010	72	34	47.2	720	193	26.8	67	38	56.7	670	159	23.7
2011	56	32	57.1	560	158	28.2	63	25	39.7	630	153	24.3
2012	86	41	47.7	860	240	27.9	83	31	37.3	830	221	26.6
2013	104	52	50.0	1040	286	27.5	95	32	33.7	950	260	27.4
2014	116	59	50.9	1160	359	30.9	82	36	43.9	820	239	29.1
2015	39	16	41.0	390	112	28.7	35	10	28.6	350	112	32.0
Skin cancer screening within the recent 3 months *before date of diagnosis**												
2009	68	31	45.6	680	33	4.9	71	21	29.6	710	23	3.2
2010	72	25	34.7	720	22	3.1	67	23	34.3	670	25	3.7
2011	56	25	44.6	560	21	3.8	63	17	27.0	630	25	4.0
2012	86	29	33.7	860	48	5.6	83	20	24.1	830	36	4.3
2013	104	36	34.6	1040	43	4.1	95	26	27.4	950	27	2.8
2014	116	37	31.9	1160	56	4.8	82	19	23.2	820	41	5.0
2015	39	14	35.9	390	12	3.1	35	7	20.0	350	15	4.3

*For controls, the pseudodiagnosis date were used; N: total number of cases or controls in the stratum; n: number of cases or controls in the stratum that had a positive screening history (back to 2008), the recent 24 months before diagnosis or the recent 3 months before diagnosis.

symptoms are clarified as early as possible [16]. The use of billing codes not only for screening without an occasion but also for early diagnosis creates a bias in SCS prevalence among cases in the case–control study that makes a valid estimate of the association between participation in SCS and melanoma mortality impossible. Nevertheless, if the biased data for SCS were used to estimate the effect of SCS on melanoma mortality,

Table 3
Skin cancer screening prevalence within the recent 3 or 24 months before date of diagnosis among skin melanoma deaths and before pseudodiagnosis date among controls among 10,370 controls among General Local Health Insurance Fund ("Allgemeine Ortskrankenkasse, AOK") in Germany, 2014–2015.

Age (Years)	Men				Women			
	Cases (n = 541)		Controls (n = 5410)		Cases (n = 496)		Controls (n = 4960)	
	N	%	N	%	N	%	N	%
Skin cancer screening within the recent 3 months *before date of diagnosis**								
Overall 35–99	197	36.4	235	4.3	133	26.8	192	3.9
35–49	8	22.2	6	1.9	10	28.6	10	2.8
50–59	15	23.8	19	2.9	13	38.2	9	3.2
60–69	38	41.3	43	4.8	13	27.1	27	5.2
70–79	84	42.0	93	4.6	45	29.4	60	3.9
80–89	47	34.6	67	4.9	44	24.4	77	4.0
90–99	5	35.7	7	5.8	8	17.4	9	2.8
Skin cancer screening within the recent 24 months *before date of diagnosis**								
Overall 35–99	273	50.5	1.440	26.6	199	40.1	1.233	24.9
35–49	13	36.1	61	19.1	16	45.7	72	20.0
50–59	27	42.9	125	18.9	17	50.0	70	25.0
60–69	50	54.3	228	25.3	18	37.5	134	25.8
70–79	113	56.5	601	29.6	70	45.8	432	27.9
80–89	63	46.3	399	28.9	66	36.7	457	23.7
90–99	7	50.0	26	21.7	12	26.1	68	21.3

*For controls, the pseudodiagnosis date was used.

SCS would be associated with an increased risk of dying from melanoma.

A mortality evaluation of SCS with health insurance data seems only possible if these data are individually merged with information from the screening physician (reason for skin examination) and with data from population-based cancer registries (exact histology, tumour stage, and cause of death), similar to what is planned in the mortality evaluation of the mammography screening program in Germany [17]. However, the assessment of whether a skin examination was actually a screening or a diagnostic work-up would depend on the validity of the information provided by the physician performing the skin examination that needs to be assessed. An alternative to a complicated pooling of these data under consideration of German data protection laws is to conduct a case–control study based on primary data collection. However, this approach is likely to introduce a substantial error in self-reported data or data from surrogates of deceased cases on previous SCS examinations that have taken place.

A major limitation of our study is the use of health insurance data. Because health insurers only record that an insured person died, we had to indirectly infer death due to melanoma using an adapted algorithm. It is likely that this derivation was accompanied by errors even though we chose stringent inclusion criteria to reduce misclassification of deaths, which potentially lowered the expected number of eligible melanoma deaths by 650 cases. Furthermore, we were unable to characterise diagnosed cutaneous malignant melanomas in more detail in terms of exact histology, location, and tumour stage because this information is not available in health insurance data.

In summary, a mortality evaluation of the SCS cannot be validly performed on the basis of health insurance data alone, so that the question of the influence of the SCS on melanoma mortality remains unanswered. The record linkage of different data sources described above will take several years, even if the project is carried out quickly. Therefore, other evaluation outcomes than mortality of the SCS must be used to assess the benefits and harms of the SCS [18]. These surrogate outcomes include the decrease in melanoma incidence of high stages, the decrease in aggressiveness of treatment, the decrease in disease-related psychological distress, and the decrease in disease-related economic costs.

Author contributions

AS: conceptualised the study and developed the methodology, was involved in the programming, prepared the initial draft, and acquired the financial support for the project leading to this publication; KS: developed the methodology, verified study results, produced metadata, critically reviewed and commented on the initial draft, and coordinated the project; PT: developed the methodology, programmed data analyses, and critically reviewed and commented on the initial draft; .SN: developed the methodology, critically reviewed and commented on the initial draft; CS: developed the methodology, critically reviewed and commented on the initial draft; KP: developed the methodology, critically reviewed and commented on the initial draft; JW: developed the methodology, critically reviewed and commented on the initial draft, and acquired the financial support for the project leading to this publication; HL: developed the methodology, critically reviewed and commented on the initial draft; MN: developed the methodology, critically reviewed and commented on the initial draft.

Conflict of interest statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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